

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 4 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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Mahidol University
Faculty of Information
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ITDS261 Introduction to Software Engineering

Lecture 6 – Requirements modeling with flow modeling I

Recap

- Requirements Engineering
  - User requirements
  - System requirements
    - Function requirements
    - Non-functional requirements
- Use case modeling
  - Use case diagrams
  - Use case narrative



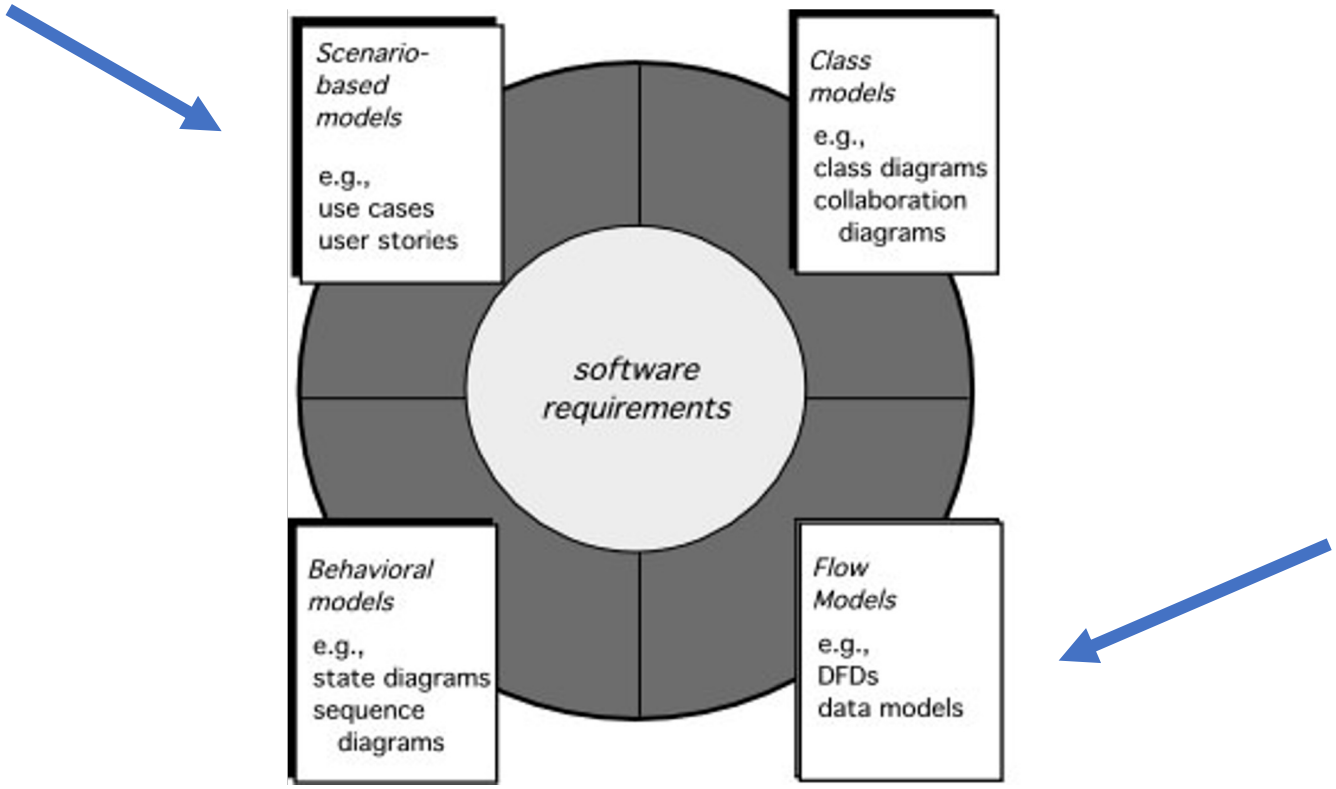
Topics

- Flow-oriented modeling
- Model consistency checking

Requirements Modeling Strategies

- One view of requirements modeling, called *structured analysis*, considers data and the processes that transform the data as separate entities
  - **Data objects** are modeled in a way that defines their attributes and relationships
  - **Processes** that manipulate data objects are modeled in a manner that shows how they transform data as data objects flow through the system
- A second approach to analysis modeled, called *object-oriented analysis*, focuses on
  - the definition of classes and
  - the manner in which they collaborate with one another reflect customer requirements

Elements of Requirements Analysis



Flow-Oriented Modeling

- Represents how data objects are transformed as they move through the system
- Data flow diagram (DFD) is the diagrammatic form that is used
- Considered by many to be an "old school" approach, but continues to provide a view of the system that is unique—it should be used to supplement other analysis model elements

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The Flow Model

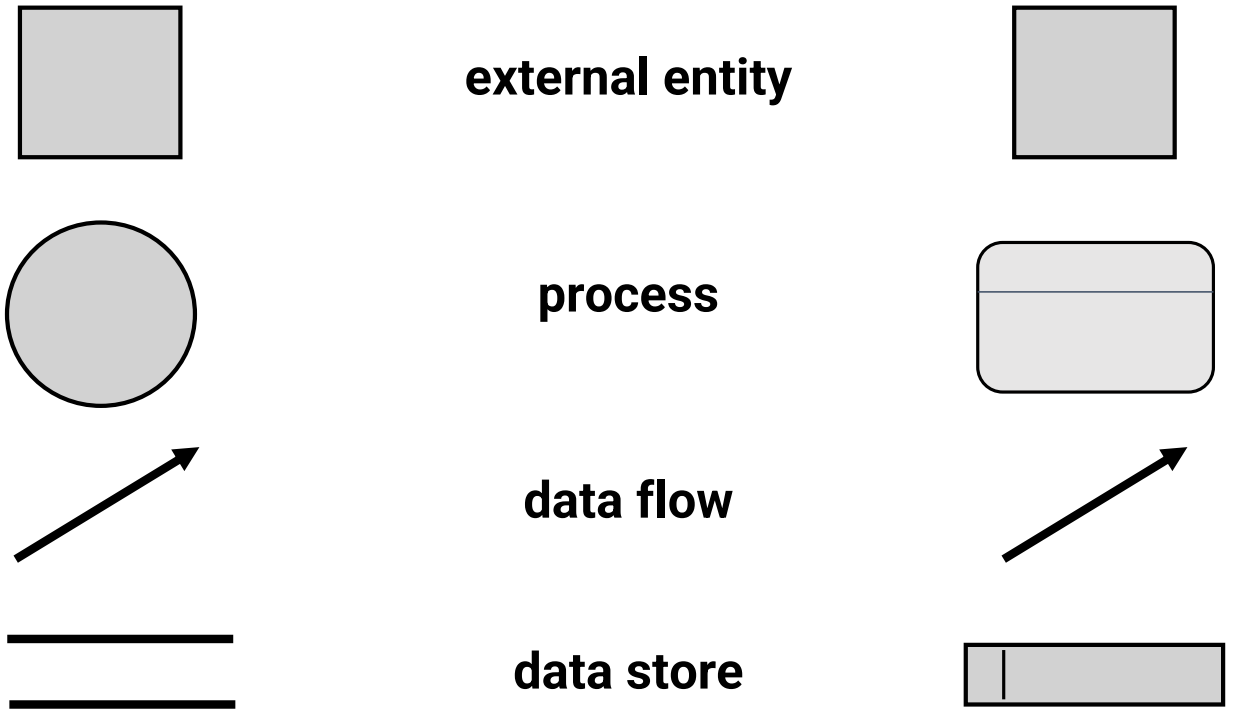
Every computer-based system transforms ....



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Data Flow Modeling

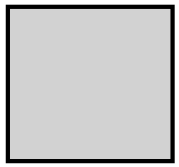
Flow Modeling Notation

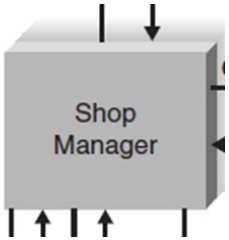


Yourdon/DeMarco's

Gane/Sarson's

External Entity





A producer or consumer of data

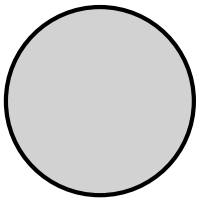
A person, organization, or system that is external to the system

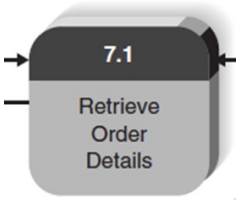
Has interactions with the system (adds data to system or receives data from system)

Examples: a person, a device, a sensor, a computer-based system

Data must always originate somewhere and must always be sent to something

Process





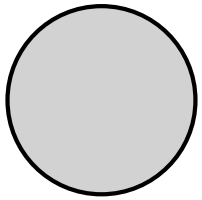
A data transformer
(changes input to output)

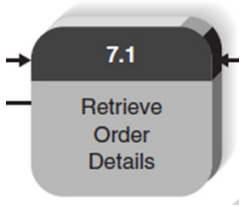
An activity or function where the manipulation and transformation of data takes place

Examples: *compute taxes, determine area, format report, display graph*

Data must always be processed in some way to achieve system function

Process





You do include logical processes that:

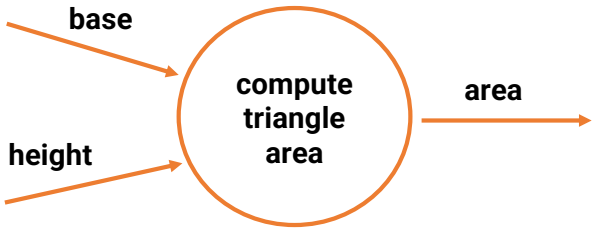
- ***Perform computations*** (e.g., calculate grade point average)
- ***Make decisions*** (e.g., determine availability of ordered products)
- ***Sort, filter or otherwise summarize data*** (e.g., identify overdue invoices)
- ***Organize data*** into useful information (e.g., generate a report or answer a question)
- ***Trigger other processes*** (e.g., turn on the furnace or instruct a robot)
- ***Use stored data*** (create, read, update or delete a record)

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Data Flow



Data flows through a system, beginning as input and transformed into output



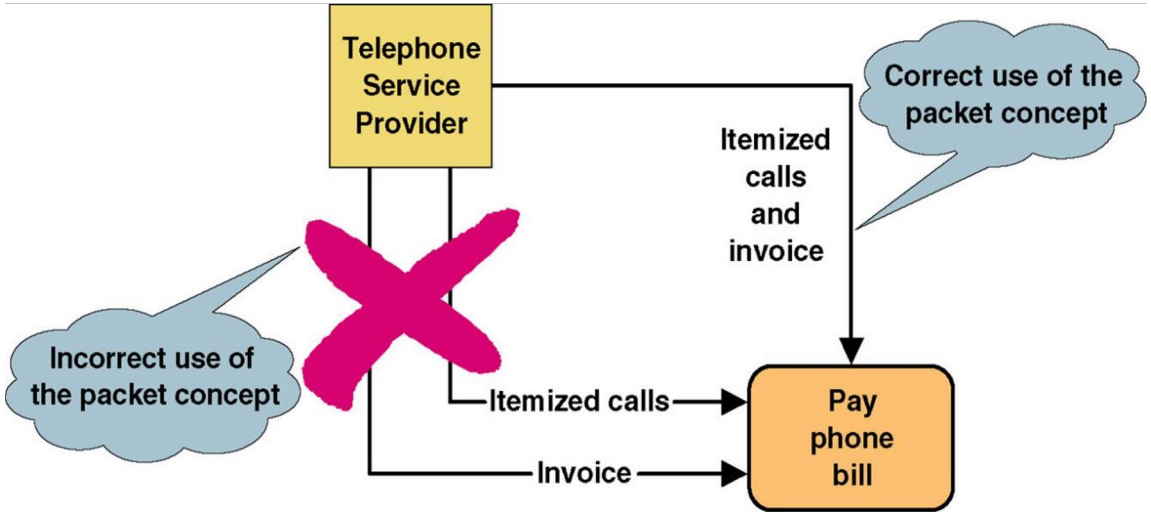
A data flow is *data in motion.*

An **input of data to a process**, or **the output of data** (or information) **from a process.**

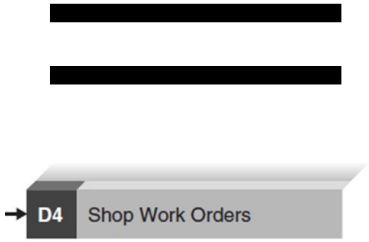
The **creation, deletion, or update of data in a file or database** (called a data store on the DFD).

Data Flow Packet Concept

- Data that should travel together should be shown as a single data flow, no matter how many physical documents might be included



Data Store

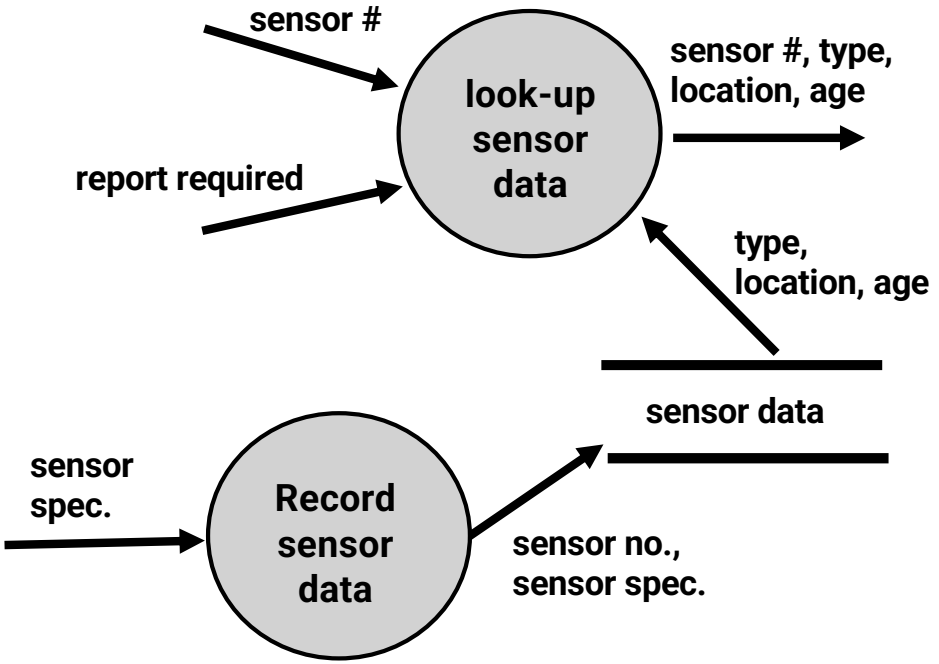


The storage of persistent data required and/or produced by the process

Data flows **leaving** the data store are *data retrievals*

Data flows **entering** the data store are *updates or new data added*

Data is often stored for later use



Depicting Business Processes with DFDs

- Business processes are too complex to be shown on a single DFD
- A deliberate hierarchy is created with multiple “levels” of DFDs
- To build the hierarchy, use ***Decomposition***
  - Child diagrams show a portion of the parent diagram in greater detail

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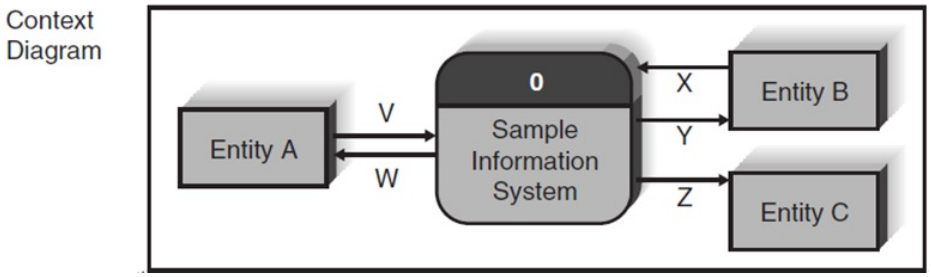
Data Flow Diagramming: Guidelines

- All icons must be labeled with meaningful names
- The DFD evolves through a number of levels of detail
- Always begin with a context level diagram
  (also called level 0)
- Always show external entities at level 0
- Always label data flow arrows

Constructing a DFD

- Review user scenarios and/or the data model to isolate data objects and use a grammatical parse to determine “operations”
- Determine external entities (producers and consumers of data)
- Create a level 0 DFD (i.e., context diagram)

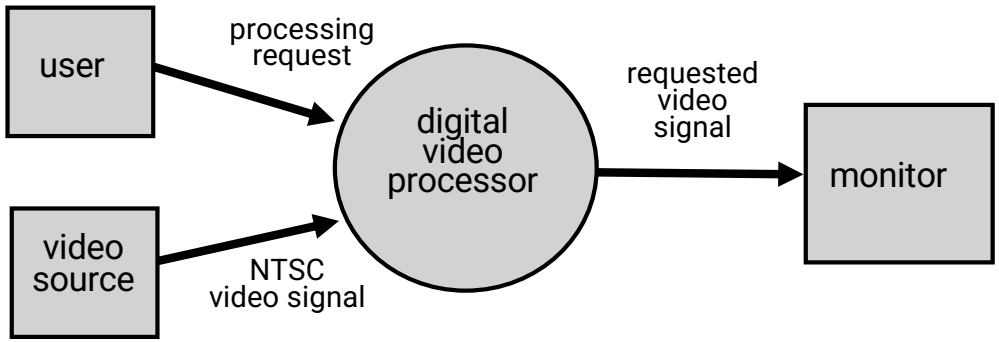
Context Diagram



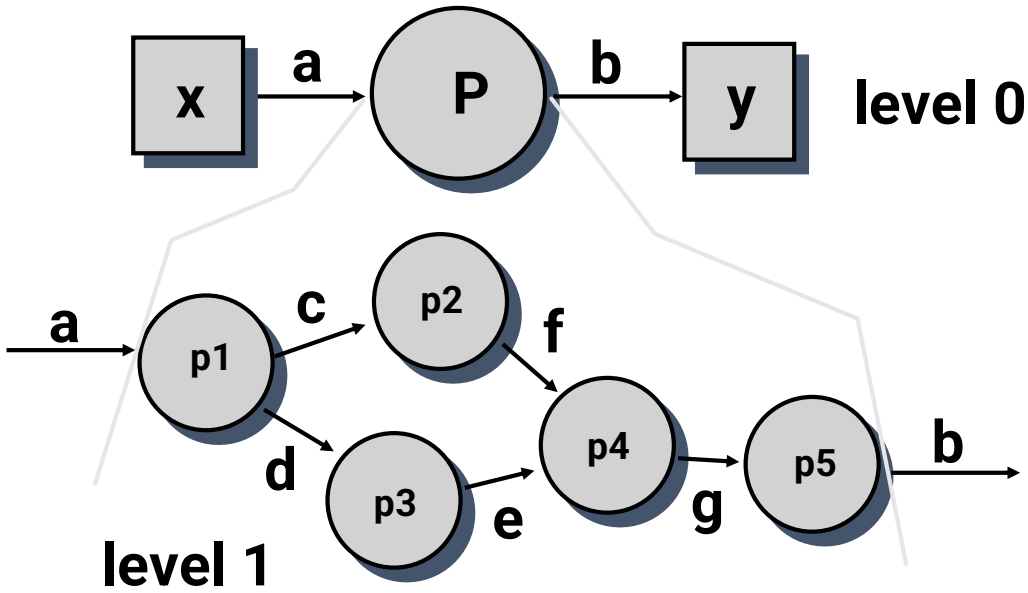
- Top-level DFD in every process model
- Shows the context into which the business process fits
- Shows the overall business process as just one process (process ‘zero’)
- Shows all the external entities that receive information from or contribute information to the system

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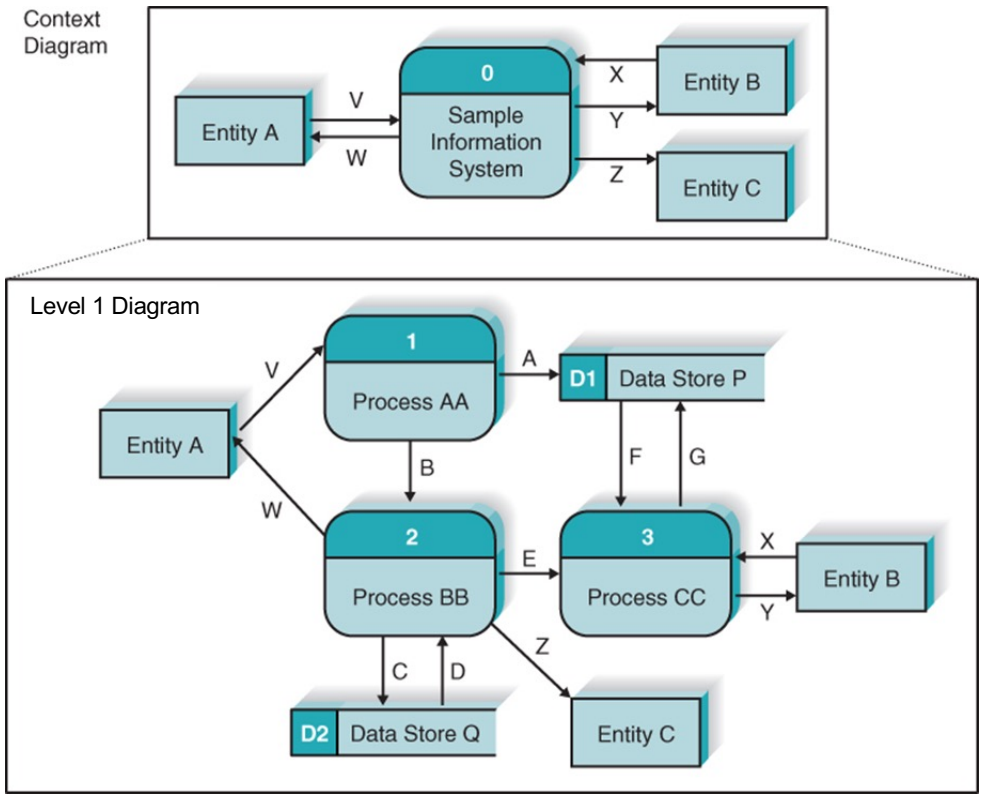
Level-0 DFD (context diagram): Example



The Data Flow Hierarchy



The Data Flow Hierarchy



Flow Modeling Notes

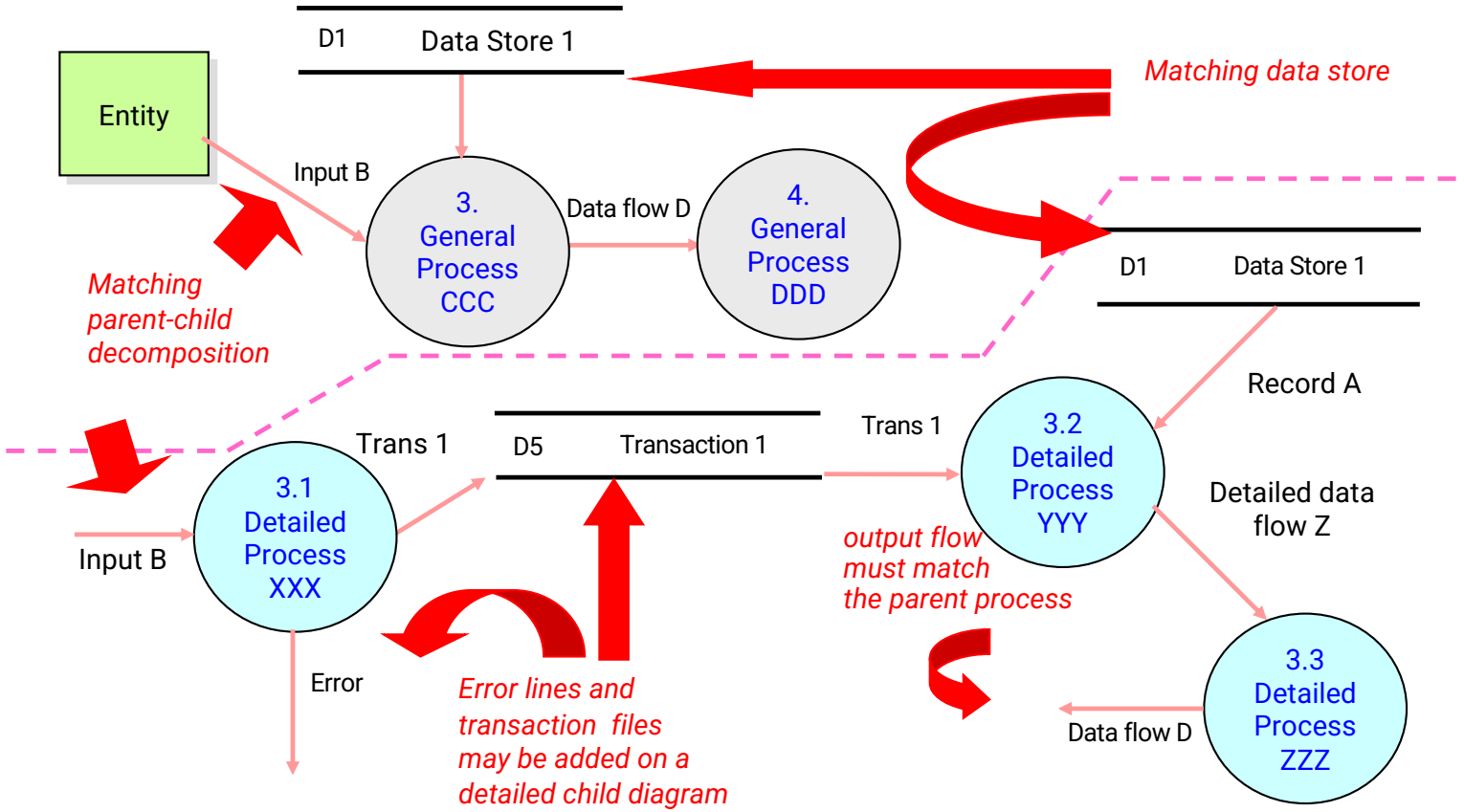
- Each bubble is refined until it does just one thing
- The expansion ratio decreases as the number of levels increase
- Most systems require between 3 and 7 levels for an adequate flow model
- A single data flow item (arrow) may be expanded as levels increase (data dictionary provides information)

Balancing

- Ensures that information presented at one level of a DFD is accurately represented in the next level DFD.
- Data flows on parent diagram are carried down to child diagram.
- Child diagram adds new processes and new data flows

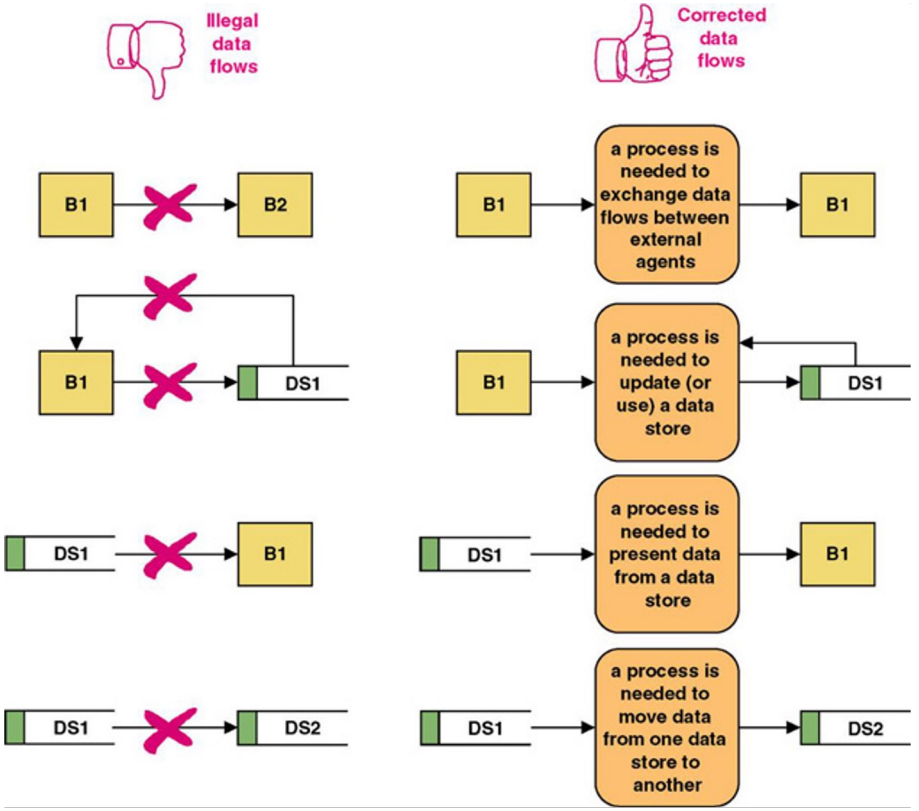
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Differences between the Parent Diagram and the Child Diagram



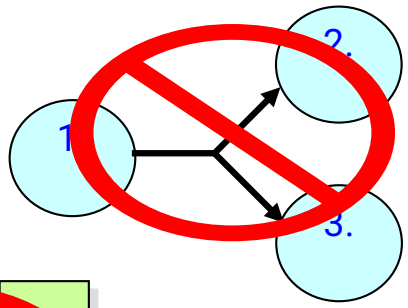
Rules for Data Flows

- A data flow should never go unnamed
- In logical modeling, data flow names should describe the data flow without describing the implementation
- All data flows must begin and/or end at a process

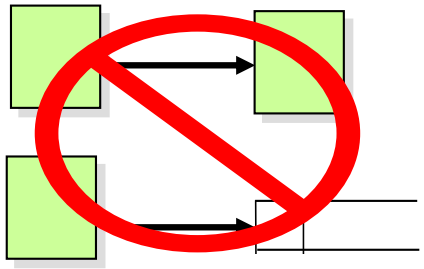


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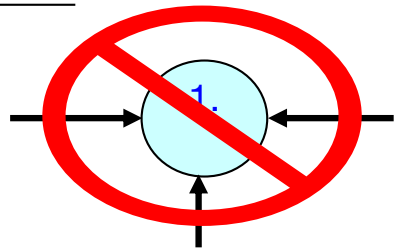
Don’t Break DFD Rules!



Data flows should not split into two or more different data flows



All data flows must EITHER originate or terminate at a process



Processes need to have at least one input data flow and one output data flow

Dataflow Analysis: Internal Consistency

- Context Diagram
  - Must exist
  - Must contain only one process
  - At least one input from an external entity and one output to an external entity
  - The process must be numbered 0
- Detailed DFDs
  - At least one process
  - No more than seven processes
- Process
  - At least one input data flow and one output data flow
  - Must be connected to a data store, process, or external entity
  - Must be labeled

Dataflow Analysis: Internal Consistency (cont.)

- External Entity
  - Must only appear on the context diagram
  - Must be connected to a process
  - Must be labeled
- Data Flow
  - Must be an interface between a process and either a second process, a data store, or an external entity
  - A data flow into (from) a data store must have a composition that is a subset of the data store's composition
  - Must be labeled
- Data Store
  - Must be an interface between two processes
  - Must be labeled

Dataflow Analysis: Hierarchical Consistency

- Data Flow Diagram
  - A parent diagram must exist unless it is a context diagram
- Process
  - Must decompose to either another diagram or a primitive process specification
  - Must be numbered with respect to its parent
- Data Flow
  - An input (output) data flow on a parent diagram must appear on a child diagram as input (output)
  - An input (output) data flow on a child diagram must appear on a parent diagram as input (output)

Dataflow Analysis: Hierarchical Consistency (cont.)

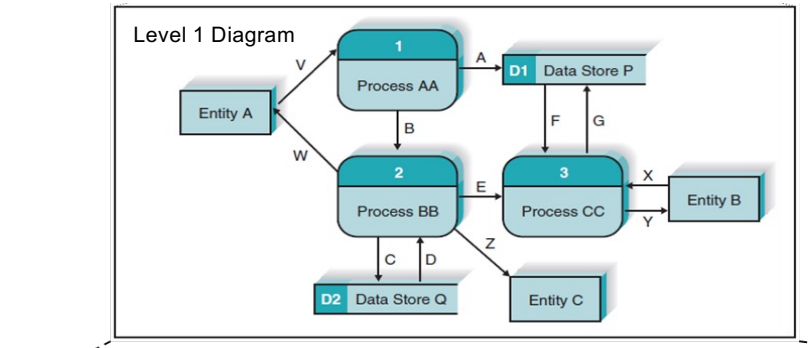
- Data Flow (cont.)
  - A set of data flows on a child diagram that were split from a data flow on a parent diagram must match the parent data flow's composition
  - A data flow must decompose to either a record definition of an element definition *(can be one in ERD)*
- Data Store
  - A data store must decompose to either a file definition or a record definition *(can be one in ERD)*
- Primitive Process Specification
  - All inputs and outputs must match those of the corresponding primitive process on the data flow diagram
  - Must be labeled with the same identifier as the corresponding primitive process on the data flow diagram

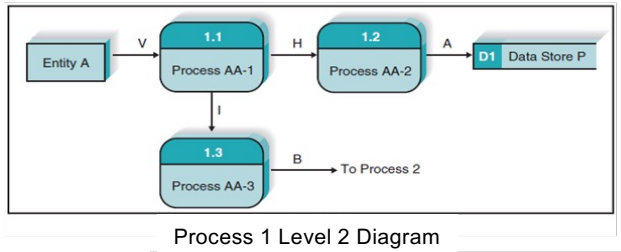
Diagram Numbering

- Correctly numbering each process helps the user understand where the process fits into the overall hierarchy
  - Context Diagram is always “Level 0”
  - Level 1 processes are always numbered with integer value (1, 2, 3, etc.)
  - Level 2 processes always have one “dot”: parent number “dot” unique number
    (1.1, 1.2, 1.3, etc.)
  - Level 3 processes always have two “dots”: parent number “dot” unique number
    (1.1.1, 1.1.2, 1.1.3, etc.)

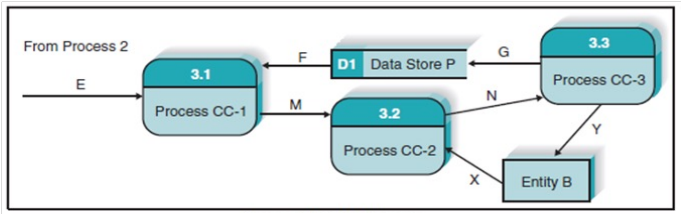
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DFD Numbering + Hierarchical Consistency

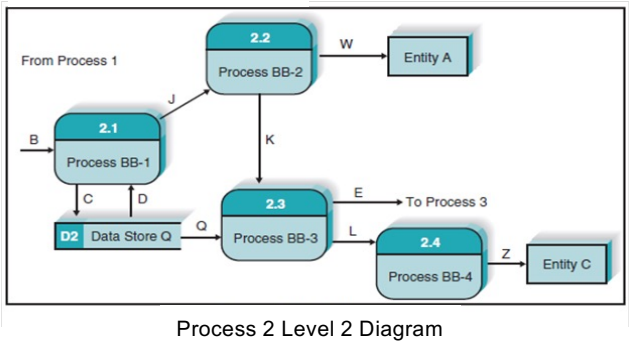




Process 1 Level 2 Diagram



Process 3 Level 2 Diagram

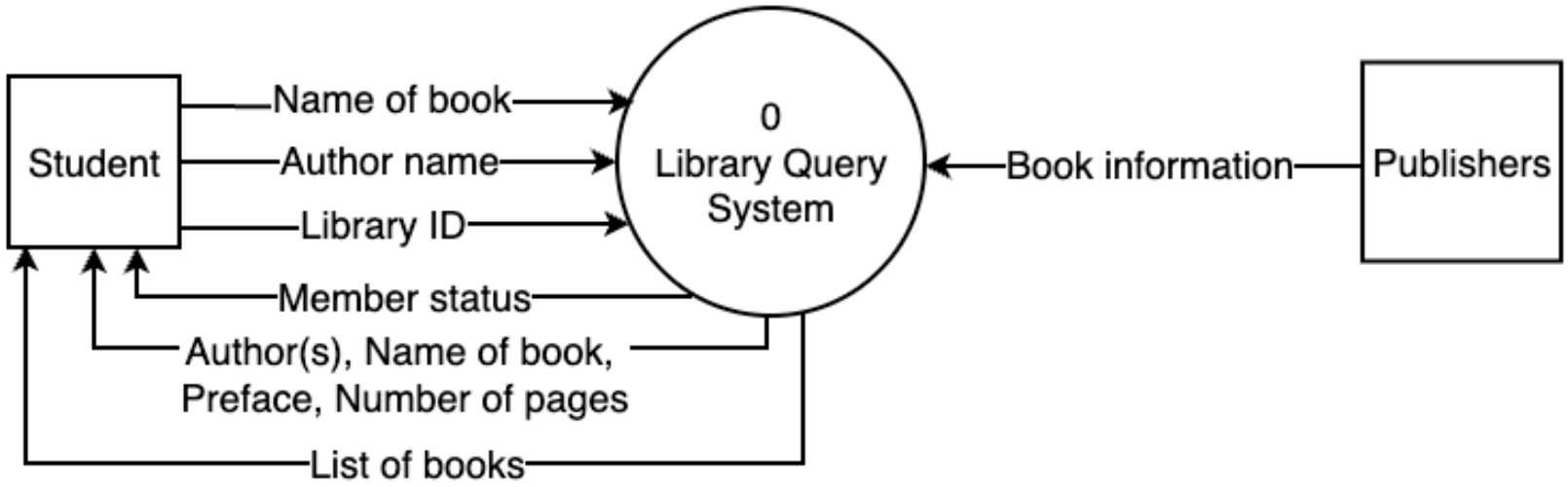


Process 2 Level 2 Diagram

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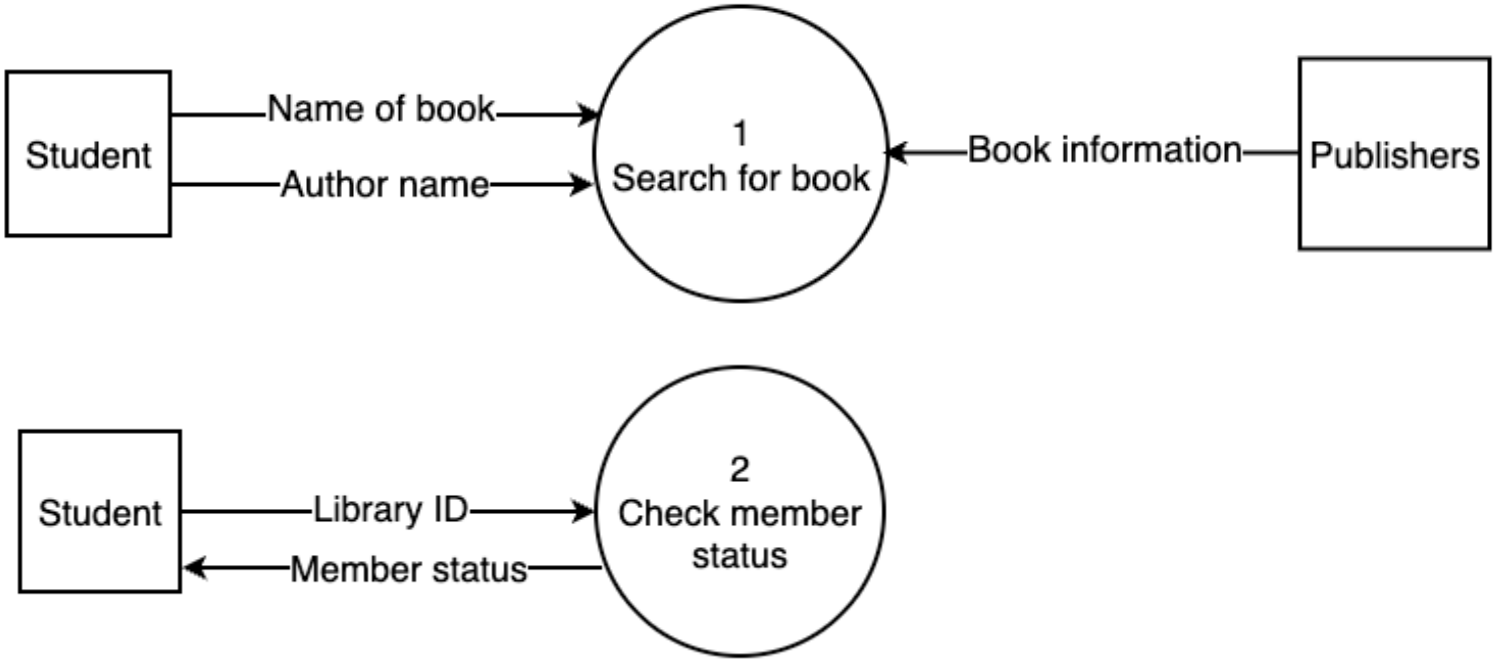
Example: DFD for Library Query System

Level 0



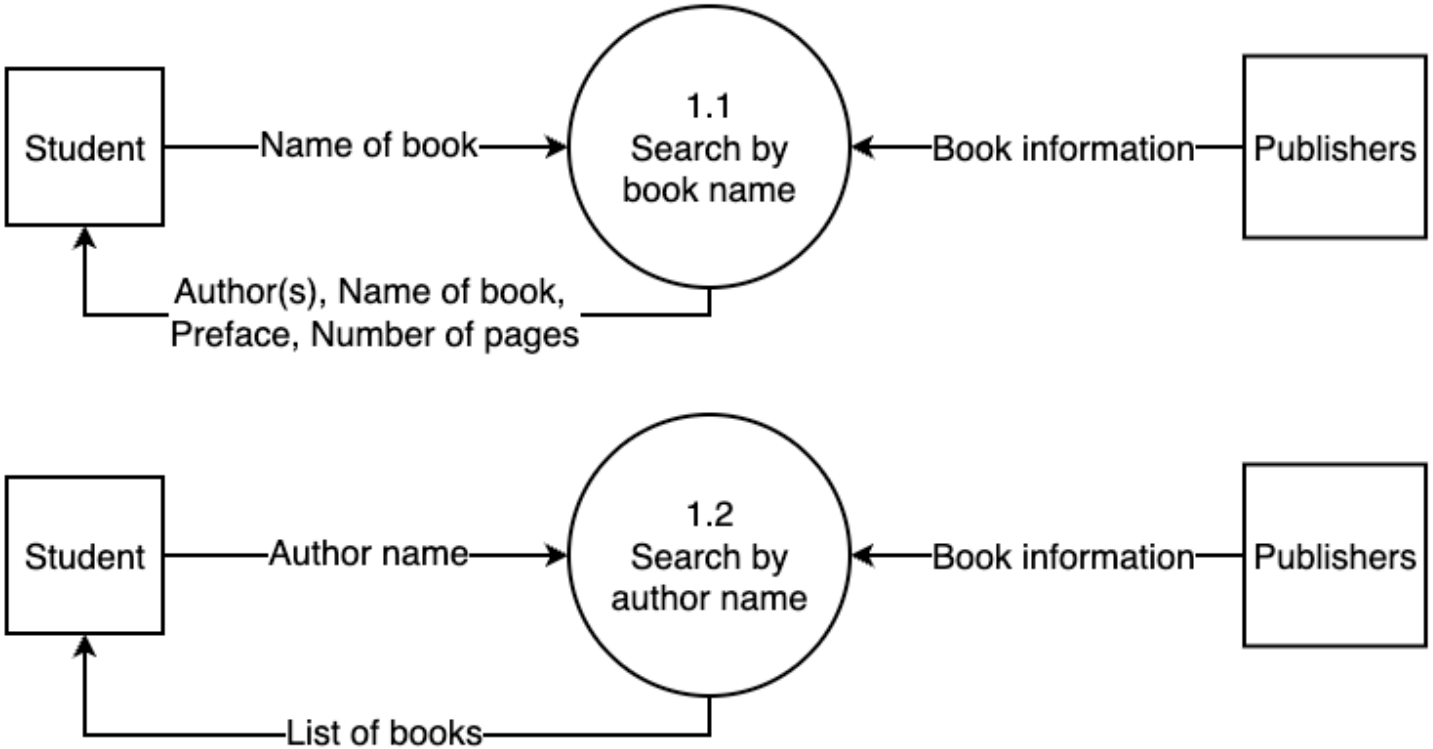
Example: DFD for Library Query System

Level 1

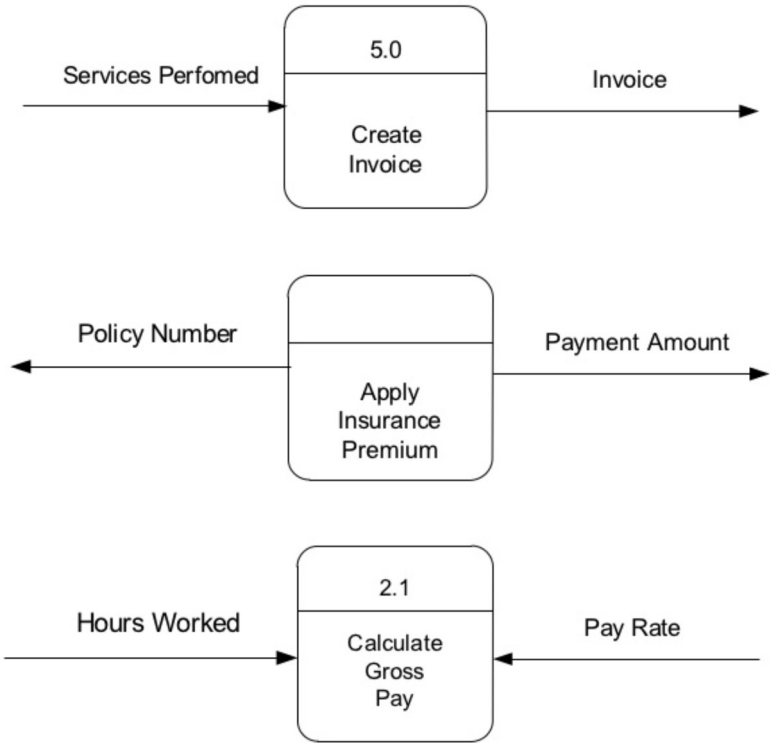


Example: DFD for Library Query System

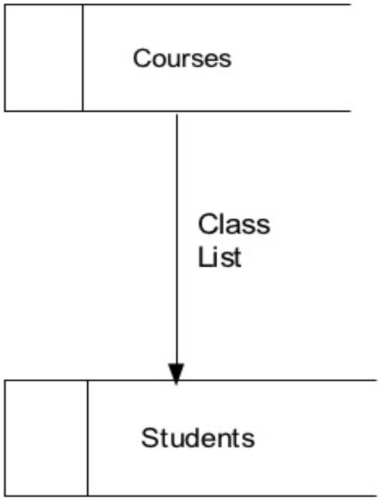
Level 2

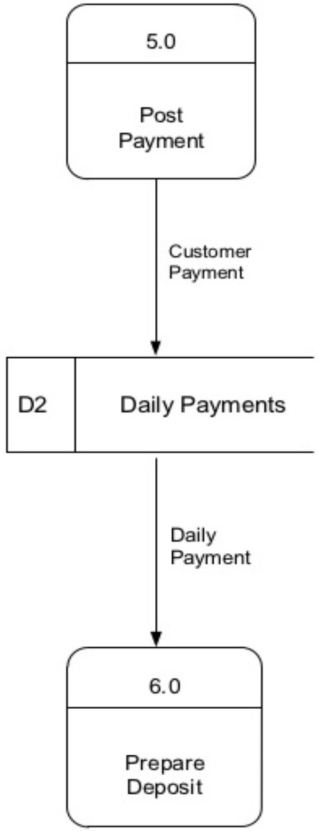


Correct/Incorrect?

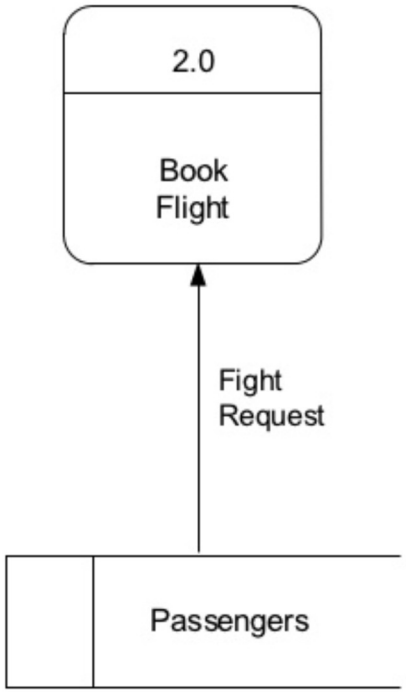


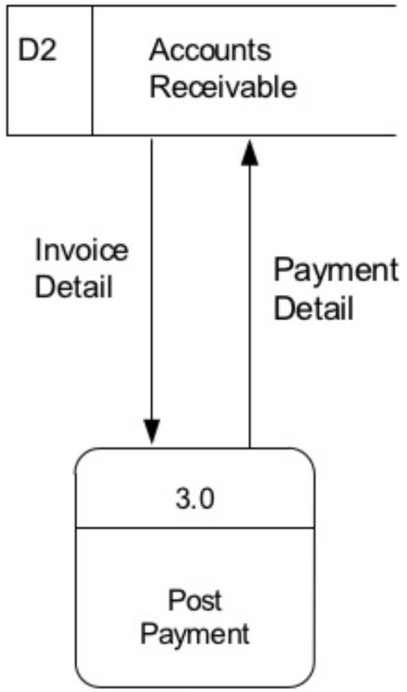
Correct/Incorrect?



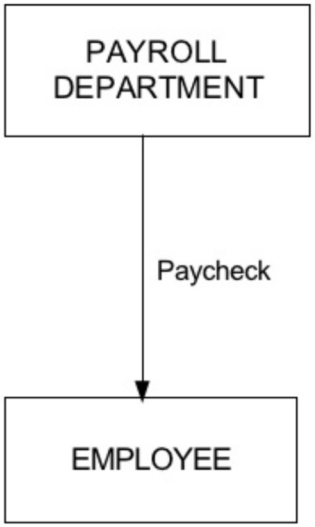


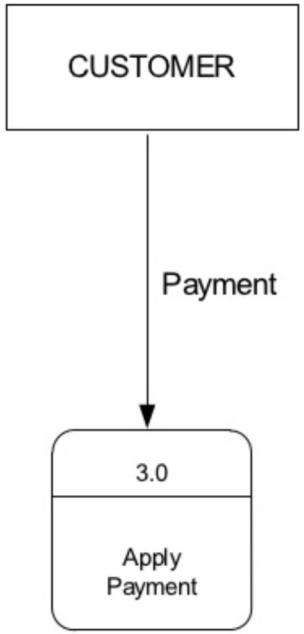
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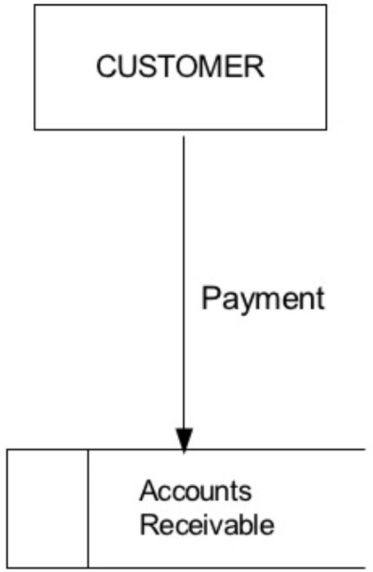




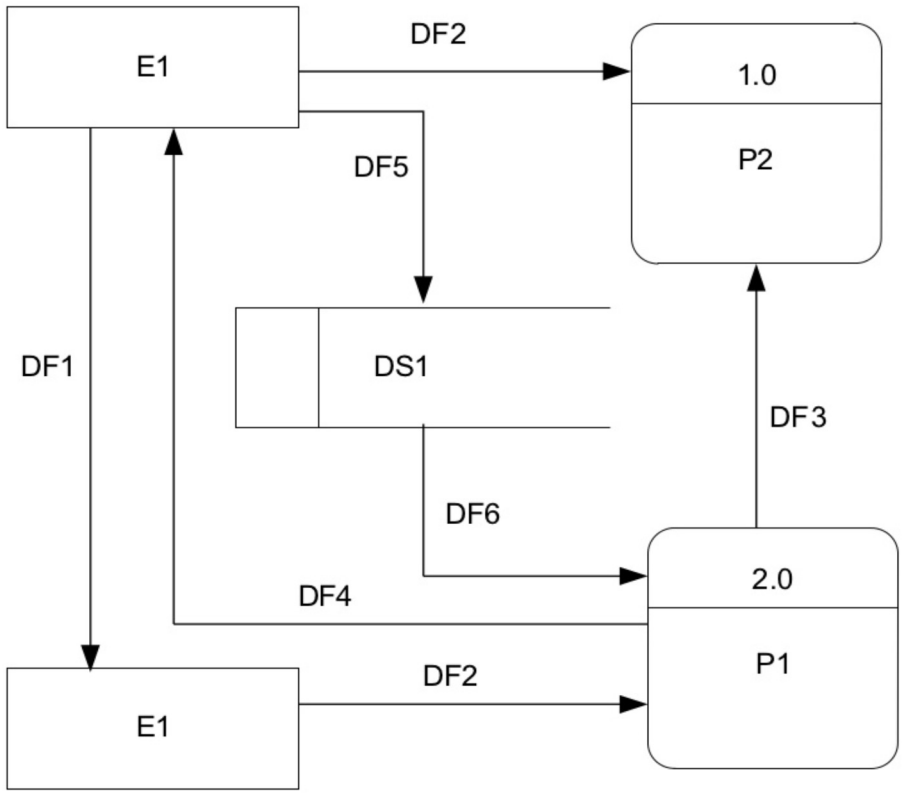
Correct/Incorrect?







List the Errors in This DFD



Exercise

- วาด context DFD ของระบบจำหน่ายบัตรเข้าสวนสัตว์ สำหรับ**ผู้เยี่ยมชม**

ผู้เยี่ยมชม สามารถซื้อตั๋วผ่านระบบโดยเลือกประเภทบัตร เช่น เด็ก ผู้ใหญ่ หรือผู้สูงอายุ และเลือกวันที่เข้าชม ระบบจะแสดงราคาตั๋วตามข้อมูลที่ผู้ดูแลระบบกำหนด จากนั้นผู้เยี่ยมชมสามารถชำระเงินผ่านบัตรเครดิตหรือ QR Code เมื่อชำระเงินสำเร็จ ระบบจะออกตั๋วพร้อม Booking ID และ QR Code ให้แก่ผู้เยี่ยมชม ผู้เยี่ยมชมสามารถตรวจสอบรายละเอียดและสถานะของตั๋วได้ หากต้องการยกเลิกตั๋วก่อนถึงวันที่เข้าชม ผู้เยี่ยมชมสามารถส่งคำขอยกเลิกผ่านระบบ โดยระบบจะตรวจสอบเงื่อนไขการคืนเงินและดำเนินการคืนเงินหากเป็นไปตามนโยบายที่กำหนด